

# 74AVC4T774

## 4-bit dual supply translating transceiver; 3-state

Rev. 1 — 25 September 2017

Product data sheet

## 1 General description

The 74AVC4T774 is a 4-bit, dual supply transceiver that enables bidirectional level translation. It features eight 1-bit input-output ports ( $A_n$  and  $B_n$ ), four direction control inputs (DIR1, DIR2, DIR3 and DIR4), an output enable input ( $\overline{OE}$ ) and dual supply pins ( $V_{CC(A)}$  and  $V_{CC(B)}$ ). Both  $V_{CC(A)}$  and  $V_{CC(B)}$  can be supplied at any voltage between 0.8 V and 3.6 V making the device suitable for translating between any of the low voltage nodes (0.8 V, 1.2 V, 1.5 V, 1.8 V, 2.5 V and 3.3 V). Pins  $A_n$ ,  $\overline{OE}$  and DIRn are referenced to  $V_{CC(A)}$  and pins  $B_n$  are referenced to  $V_{CC(B)}$ . A HIGH on DIRn allows transmission from  $A_n$  to  $B_n$  and a LOW on DIRn allows transmission from  $B_n$  to  $A_n$ . The output enable input ( $\overline{OE}$ ) can be used to disable the outputs so the buses are effectively isolated.

The device is fully specified for partial power-down applications using  $I_{OFF}$ . The  $I_{OFF}$  circuitry disables the output, preventing any damaging backflow current through the device when it is powered down. In suspend mode when either  $V_{CC(A)}$  or  $V_{CC(B)}$  are at GND level, both  $A_n$  and  $B_n$  are in the high-impedance OFF-state.

## 2 Features and benefits

- Wide supply voltage range:
  - $V_{CC(A)}$ : 0.8 V to 3.6 V
  - $V_{CC(B)}$ : 0.8 V to 3.6 V
- Complies with JEDEC standards:
  - JESD8-12 (0.8 V to 1.3 V)
  - JESD8-11 (0.9 V to 1.65 V)
  - JESD8-7 (1.2 V to 1.95 V)
  - JESD8-5 (1.8 V to 2.7 V)
  - JESD8-B (2.7 V to 3.6 V)
- ESD protection:
  - HBM JESD22-A114E Class 3B exceeds 8000 V
  - CDM JESD22-C101C exceeds 1500 V
- Maximum data rates:
  - 380 Mbit/s ( $\geq 1.8$  V to 3.3 V translation)
  - 200 Mbit/s ( $\geq 1.1$  V to 3.3 V translation)
  - 200 Mbit/s ( $\geq 1.1$  V to 2.5 V translation)
  - 200 Mbit/s ( $\geq 1.1$  V to 1.8 V translation)
  - 150 Mbit/s ( $\geq 1.1$  V to 1.5 V translation)
  - 100 Mbit/s ( $\geq 1.1$  V to 1.2 V translation)
- Suspend mode
- Latch-up performance exceeds 100 mA per JESD 78 Class II
- Inputs accept voltages up to 3.6 V
- $I_{OFF}$  circuitry provides partial Power-down mode operation

- Specified from -40 °C to +85 °C and -40 °C to +125 °C

3 Ordering information

Table 1. Ordering information

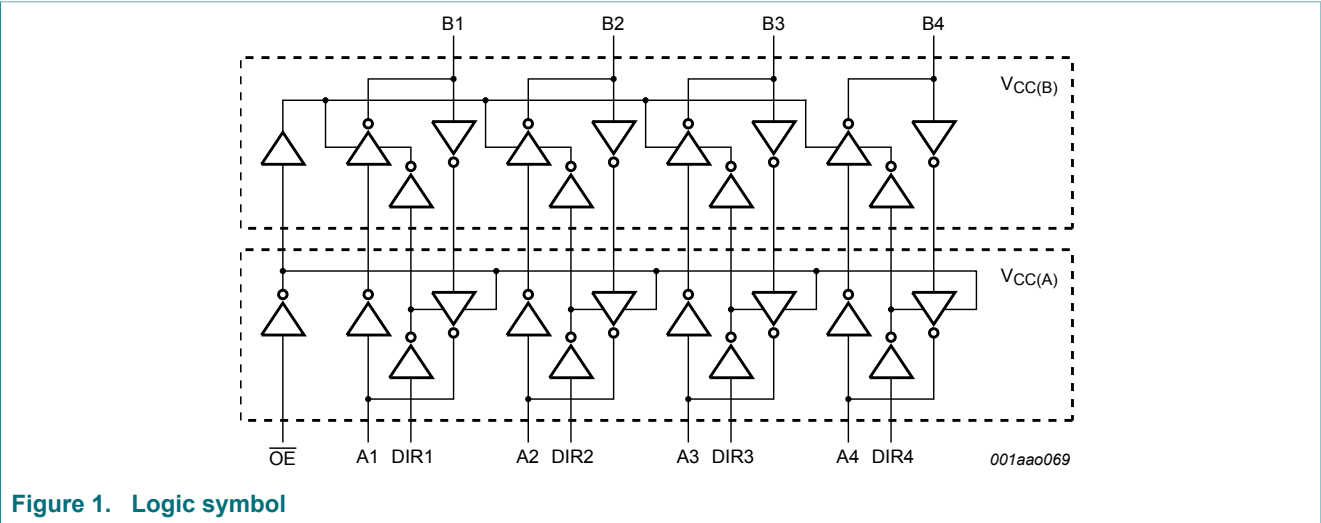
| Type number  | Package           |          |                                                                                                                                |           |
|--------------|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------|-----------|
|              | Temperature range | Name     | Description                                                                                                                    | Version   |
| 74AVC4T774BQ | -40 °C to +125 °C | DHVQFN16 | plastic dual in-line compatible thermal enhanced very thin quad flat package; no leads; 16 terminals; body 2.5 x 3.5 x 0.85 mm | SOT763-1  |
| 74AVC4T774GU | -40 °C to +125 °C | XQFN16   | plastic, extremely thin quad flat package; no leads; 16 terminals; body 1.80 x 2.60 x 0.50 mm                                  | SOT1161-1 |

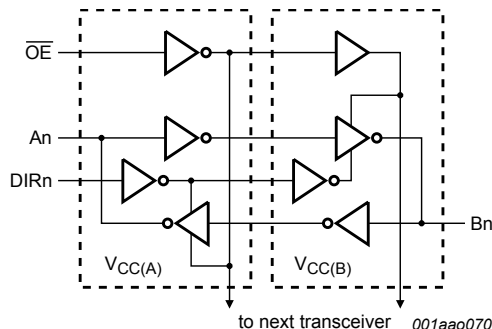
4 Marking

Table 2. Marking codes

| Type number  | Marking code |
|--------------|--------------|
| 74AVC4T774BQ | C4T774       |
| 74AVC4T774GU | B77          |

5 Functional diagram

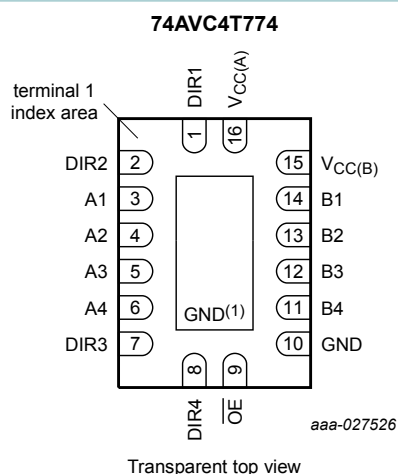




**Figure 2. Logic diagram (one 1-bit transceiver)**

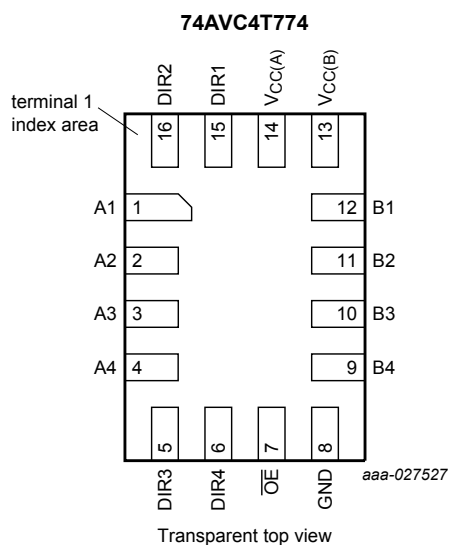
## 6 Pinning information

## 6.1 Pinning



(1) This is not a supply pin, the substrate is attached to this pad using conductive die attach material. There is no electrical or mechanical requirement to solder this pad. However if it is soldered the solder land should remain floating or be connected to GND.

### Figure 3. Pin configuration SOT763-1 (DHVQFN16)



**Figure 4. Pin configuration SOT1161-1 (XQFN16)**

## 6.2 Pin description

Table 3. Pin description

| Symbol                 | Pin            |               | Description                                                                           |
|------------------------|----------------|---------------|---------------------------------------------------------------------------------------|
|                        | SOT763-1       | SOT1161-1     |                                                                                       |
| $V_{CC(A)}$            | 16             | 14            | supply voltage A (An, $\overline{OE}$ and DIRn inputs are referenced to $V_{CC(A)}$ ) |
| DIR1, DIR2, DIR3, DIR4 | 1, 2, 7, 8     | 15, 16, 5, 6  | direction control input                                                               |
| A1, A2, A3, A4         | 3, 4, 5, 6     | 1, 2, 3, 4    | data input or output                                                                  |
| GND                    | 10             | 8             | ground (0 V)                                                                          |
| B1, B2, B3, B4         | 14, 13, 12, 11 | 12, 11, 10, 9 | data input or output                                                                  |
| $\overline{OE}$        | 9              | 7             | output enable input (active LOW)                                                      |
| $V_{CC(B)}$            | 15             | 13            | supply voltage B (Bn pins are referenced to $V_{CC(B)}$ )                             |

## 7 Functional description

Table 4. Function table <sup>[1] [2]</sup>

| Supply voltage            | Input           |      |      |      |      | Input/output |          |
|---------------------------|-----------------|------|------|------|------|--------------|----------|
| $V_{CC(A)}$ , $V_{CC(B)}$ | $\overline{OE}$ | DIR1 | DIR2 | DIR3 | DIR4 | An           | Bn       |
| 0.8 V to 3.6 V            | L               | L    | X    | X    | X    | A1 = B1      | input B1 |
| 0.8 V to 3.6 V            | L               | H    | X    | X    | X    | input A1     | B1 = A1  |
| 0.8 V to 3.6 V            | L               | X    | L    | X    | X    | A2 = B2      | input B2 |
| 0.8 V to 3.6 V            | L               | X    | H    | X    | X    | input A2     | B2 = A2  |
| 0.8 V to 3.6 V            | L               | X    | X    | L    | X    | A3 = B3      | input B3 |
| 0.8 V to 3.6 V            | L               | X    | X    | H    | X    | input A3     | B3 = A3  |
| 0.8 V to 3.6 V            | L               | X    | X    | X    | L    | A4 = B4      | input B4 |
| 0.8 V to 3.6 V            | L               | X    | X    | X    | H    | input A4     | B4 = A4  |
| 0.8 V to 3.6 V            | H               | X    | X    | X    | X    | Z            | Z        |
| GND <sup>[3]</sup>        | X               | X    | X    | X    | X    | Z            | Z        |

[1] H = HIGH voltage level; L = LOW voltage level; X = don't care; Z = high-impedance OFF-state.

[2] The An, DIRn and  $\overline{OE}$  input circuit is referenced to  $V_{CC(A)}$ ; The Bn input circuit is referenced to  $V_{CC(B)}$ .

[3] If at least one of  $V_{CC(A)}$  or  $V_{CC(B)}$  is at GND level, the device goes into suspend mode.

## 8 Limiting values

**Table 5. Limiting values**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND (ground = 0 V).

| Symbol      | Parameter               | Conditions                    | Min  | Max             | Unit |
|-------------|-------------------------|-------------------------------|------|-----------------|------|
| $V_{CC(A)}$ | supply voltage A        |                               | -0.5 | +4.6            | V    |
| $V_{CC(B)}$ | supply voltage B        |                               | -0.5 | +4.6            | V    |
| $I_{IK}$    | input clamping current  | $V_I < 0$ V                   | -50  | -               | mA   |
| $V_I$       | input voltage           | [1]                           | -0.5 | +4.6            | V    |
| $I_{OK}$    | output clamping current | $V_O < 0$ V                   | -50  | -               | mA   |
| $V_O$       | output voltage          | Active mode [1] [2] [3]       | -0.5 | $V_{CCO} + 0.5$ | V    |
|             |                         | Suspend or 3-state mode [1]   | -0.5 | +4.6            | V    |
| $I_O$       | output current          | $V_O = 0$ V to $V_{CCO}$ [2]  | -    | $\pm 50$        | mA   |
| $I_{CC}$    | supply current          | $I_{CC(A)}$ or $I_{CC(B)}$    | -    | 100             | mA   |
| $I_{GND}$   | ground current          |                               | -100 | -               | mA   |
| $T_{stg}$   | storage temperature     |                               | -65  | +150            | °C   |
| $P_{tot}$   | total power dissipation | $T_{amb} = -40$ °C to +125 °C |      |                 |      |
|             |                         | DHVQFN16 [4]                  | -    | 500             | mW   |
|             |                         | XQFN16                        | -    | 250             | mW   |

[1] The minimum input voltage ratings and output voltage ratings may be exceeded if the input and output current ratings are observed.

[2]  $V_{CCO}$  is the supply voltage associated with the output port.

[3]  $V_{CCO} + 0.5$  V should not exceed 4.6 V.

[4] For DHVQFN16 package: above 60 °C the value of  $P_{tot}$  derates linearly at 4.5 mW/K.

## 9 Recommended operating conditions

**Table 6. Recommended operating conditions**

| Symbol              | Parameter                           | Conditions                     | Min | Max       | Unit |
|---------------------|-------------------------------------|--------------------------------|-----|-----------|------|
| $V_{CC(A)}$         | supply voltage A                    |                                | 0.8 | 3.6       | V    |
| $V_{CC(B)}$         | supply voltage B                    |                                | 0.8 | 3.6       | V    |
| $V_I$               | input voltage                       |                                | 0   | 3.6       | V    |
| $V_O$               | output voltage                      | Active mode [1]                | 0   | $V_{CCO}$ | V    |
|                     |                                     | Suspend or 3-state mode        | 0   | 3.6       | V    |
| $T_{amb}$           | ambient temperature                 |                                | -40 | +125      | °C   |
| $\Delta t/\Delta V$ | input transition rise and fall rate | $V_{CCI} = 0.8$ V to 3.6 V [2] | -   | 10        | ns/V |

[1]  $V_{CCO}$  is the supply voltage associated with the output port.

[2]  $V_{CCI}$  is the supply voltage associated with the input port.

## 10 Static characteristics

**Table 7. Typical static characteristics at  $T_{amb} = 25\text{ °C}$  [1] [2]**

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

| Symbol    | Parameter                 | Conditions                                                                                                                         | Min | Typ         | Max        | Unit          |
|-----------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|------------|---------------|
| $V_{OH}$  | HIGH-level output voltage | $V_I = V_{IH}$ or $V_{IL}$                                                                                                         |     |             |            |               |
|           |                           | $I_O = -1.5\text{ mA}$ ; $V_{CC(A)} = V_{CC(B)} = 0.8\text{ V}$                                                                    | -   | 0.69        | -          | V             |
| $V_{OL}$  | LOW-level output voltage  | $V_I = V_{IH}$ or $V_{IL}$                                                                                                         |     |             |            |               |
|           |                           | $I_O = 1.5\text{ mA}$ ; $V_{CC(A)} = V_{CC(B)} = 0.8\text{ V}$                                                                     | -   | 0.07        | -          | V             |
| $I_I$     | input leakage current     | DIRn, $\overline{OE}$ input; $V_I = 0\text{ V}$ or $3.6\text{ V}$ ;<br>$V_{CC(A)} = V_{CC(B)} = 0.8\text{ V}$ to $3.6\text{ V}$    | -   | $\pm 0.025$ | $\pm 0.25$ | $\mu\text{A}$ |
| $I_{OZ}$  | OFF-state output current  | A or B port; $V_O = 0\text{ V}$ or $V_{CCO}$ ; $V_{CC(A)} = V_{CC(B)} = 3.6\text{ V}$ [3]                                          | -   | $\pm 0.5$   | $\pm 2.5$  | $\mu\text{A}$ |
|           |                           | suspend mode A port; $V_O = 0\text{ V}$ or $V_{CCO}$ ; $V_{CC(A)} = 3.6\text{ V}$ ;<br>$V_{CC(B)} = 0\text{ V}$ [3]                | -   | $\pm 0.5$   | $\pm 2.5$  | $\mu\text{A}$ |
|           |                           | suspend mode B port; $V_O = 0\text{ V}$ or $V_{CCO}$ ; $V_{CC(A)} = 0\text{ V}$ ;<br>$V_{CC(B)} = 3.6\text{ V}$ [3]                | -   | $\pm 0.5$   | $\pm 2.5$  | $\mu\text{A}$ |
| $I_{OFF}$ | power-off leakage current | A port; $V_I$ or $V_O = 0\text{ V}$ to $3.6\text{ V}$ ; $V_{CC(A)} = 0\text{ V}$ ;<br>$V_{CC(B)} = 0.8\text{ V}$ to $3.6\text{ V}$ | -   | $\pm 0.1$   | $\pm 1$    | $\mu\text{A}$ |
|           |                           | B port; $V_I$ or $V_O = 0\text{ V}$ to $3.6\text{ V}$ ; $V_{CC(B)} = 0\text{ V}$ ;<br>$V_{CC(A)} = 0.8\text{ V}$ to $3.6\text{ V}$ | -   | $\pm 0.1$   | $\pm 1$    | $\mu\text{A}$ |
| $C_I$     | input capacitance         | DIRn, $\overline{OE}$ input; $V_I = 0\text{ V}$ or $3.3\text{ V}$ ; $V_{CC(A)} = V_{CC(B)} = 3.3\text{ V}$                         | -   | 2.0         | -          | pF            |
| $C_{I/O}$ | input/output capacitance  | A and B port; $V_O = 3.3\text{ V}$ or $0\text{ V}$ ; $V_{CC(A)} = V_{CC(B)} = 3.3\text{ V}$                                        | -   | 4.0         | -          | pF            |

[1]  $V_{CCO}$  is the supply voltage associated with the output port.

[2]  $V_{CCI}$  is the supply voltage associated with the data input port.

[3] For I/O ports, the parameter  $I_{OZ}$  includes the input leakage current.

Table 8. Static characteristics <sup>[1] [2]</sup>

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

| Symbol          | Parameter                 | Conditions                                                                            | -40 °C to +85 °C       |                        | -40 °C to +125 °C      |                        | Unit |
|-----------------|---------------------------|---------------------------------------------------------------------------------------|------------------------|------------------------|------------------------|------------------------|------|
|                 |                           |                                                                                       | Min                    | Max                    | Min                    | Max                    |      |
| V <sub>IH</sub> | HIGH-level input voltage  | data input                                                                            |                        |                        |                        |                        |      |
|                 |                           | V <sub>CCI</sub> = 0.8 V                                                              | 0.70V <sub>CCI</sub>   | -                      | 0.70V <sub>CCI</sub>   | -                      | V    |
|                 |                           | V <sub>CCI</sub> = 1.1 V to 1.95 V                                                    | 0.65V <sub>CCI</sub>   | -                      | 0.65V <sub>CCI</sub>   | -                      | V    |
|                 |                           | V <sub>CCI</sub> = 2.3 V to 2.7 V                                                     | 1.6                    | -                      | 1.6                    | -                      | V    |
|                 |                           | V <sub>CCI</sub> = 3.0 V to 3.6 V                                                     | 2                      | -                      | 2                      | -                      | V    |
|                 |                           | DIRn, $\overline{\text{OE}}$ input                                                    |                        |                        |                        |                        |      |
|                 |                           | V <sub>CC(A)</sub> = 0.8 V                                                            | 0.70V <sub>CC(A)</sub> | -                      | 0.70V <sub>CC(A)</sub> | -                      | V    |
|                 |                           | V <sub>CC(A)</sub> = 1.1 V to 1.95 V                                                  | 0.65V <sub>CC(A)</sub> | -                      | 0.65V <sub>CC(A)</sub> | -                      | V    |
|                 |                           | V <sub>CC(A)</sub> = 2.3 V to 2.7 V                                                   | 1.6                    | -                      | 1.6                    | -                      | V    |
|                 |                           | V <sub>CC(A)</sub> = 3.0 V to 3.6 V                                                   | 2                      | -                      | 2                      | -                      | V    |
| V <sub>IL</sub> | LOW-level input voltage   | data input                                                                            |                        |                        |                        |                        |      |
|                 |                           | V <sub>CCI</sub> = 0.8 V                                                              | -                      | 0.30V <sub>CCI</sub>   | -                      | 0.30V <sub>CCI</sub>   | V    |
|                 |                           | V <sub>CCI</sub> = 1.1 V to 1.95 V                                                    | -                      | 0.35V <sub>CCI</sub>   | -                      | 0.35V <sub>CCI</sub>   | V    |
|                 |                           | V <sub>CCI</sub> = 2.3 V to 2.7 V                                                     | -                      | 0.7                    | -                      | 0.7                    | V    |
|                 |                           | V <sub>CCI</sub> = 3.0 V to 3.6 V                                                     | -                      | 0.8                    | -                      | 0.8                    | V    |
|                 |                           | DIRn, $\overline{\text{OE}}$ input                                                    |                        |                        |                        |                        |      |
|                 |                           | V <sub>CC(A)</sub> = 0.8 V                                                            | -                      | 0.30V <sub>CC(A)</sub> | -                      | 0.30V <sub>CC(A)</sub> | V    |
|                 |                           | V <sub>CC(A)</sub> = 1.1 V to 1.95 V                                                  | -                      | 0.35V <sub>CC(A)</sub> | -                      | 0.35V <sub>CC(A)</sub> | V    |
|                 |                           | V <sub>CC(A)</sub> = 2.3 V to 2.7 V                                                   | -                      | 0.7                    | -                      | 0.7                    | V    |
|                 |                           | V <sub>CC(A)</sub> = 3.0 V to 3.6 V                                                   | -                      | 0.8                    | -                      | 0.8                    | V    |
| V <sub>OH</sub> | HIGH-level output voltage | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                   |                        |                        |                        |                        |      |
|                 |                           | I <sub>O</sub> = -100 µA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 0.8 V to 3.6 V | V <sub>CCO</sub> - 0.1 | -                      | V <sub>CCO</sub> - 0.1 | -                      | V    |
|                 |                           | I <sub>O</sub> = -3 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 1.1 V            | 0.85                   | -                      | 0.85                   | -                      | V    |
|                 |                           | I <sub>O</sub> = -6 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 1.4 V            | 1.05                   | -                      | 1.05                   | -                      | V    |
|                 |                           | I <sub>O</sub> = -8 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 1.65 V           | 1.2                    | -                      | 1.2                    | -                      | V    |
|                 |                           | I <sub>O</sub> = -9 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 2.3 V            | 1.75                   | -                      | 1.75                   | -                      | V    |
|                 |                           | I <sub>O</sub> = -12 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 3.0 V           | 2.3                    | -                      | 2.3                    | -                      | V    |

| Symbol           | Parameter                 | Conditions                                                                                                                          | -40 °C to +85 °C |      | -40 °C to +125 °C |      | Unit |
|------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------|------|-------------------|------|------|
|                  |                           |                                                                                                                                     | Min              | Max  | Min               | Max  |      |
| V <sub>OL</sub>  | LOW-level output voltage  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                                                                 |                  |      |                   |      |      |
|                  |                           | I <sub>O</sub> = 100 µA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 0.8 V to 3.6 V                                                | -                | 0.1  | -                 | 0.1  | V    |
|                  |                           | I <sub>O</sub> = 3 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 1.1 V                                                           | -                | 0.25 | -                 | 0.25 | V    |
|                  |                           | I <sub>O</sub> = 6 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 1.4 V                                                           | -                | 0.35 | -                 | 0.35 | V    |
|                  |                           | I <sub>O</sub> = 8 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 1.65 V                                                          | -                | 0.45 | -                 | 0.45 | V    |
|                  |                           | I <sub>O</sub> = 9 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 2.3 V                                                           | -                | 0.55 | -                 | 0.55 | V    |
|                  |                           | I <sub>O</sub> = 12 mA;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 3.0 V                                                          | -                | 0.7  | -                 | 0.7  | V    |
| I <sub>I</sub>   | input leakage current     | DIRn, $\overline{\text{OE}}$ input; V <sub>I</sub> = 0 V or 3.6 V;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 0.8 V to 3.6 V      | -                | ±1   | -                 | ±5   | µA   |
| I <sub>OZ</sub>  | OFF-state output current  | A or B port; V <sub>O</sub> = 0 V or V <sub>CCO</sub> ;<br>V <sub>CC(A)</sub> = V <sub>CC(B)</sub> = 3.6 V <sup>[3]</sup>           | -                | ±5   | -                 | ±30  | µA   |
|                  |                           | suspend mode A port; V <sub>O</sub> = 0 V or V <sub>CCO</sub> ; V <sub>CC(A)</sub> = 3.6 V; V <sub>CC(B)</sub> = 0 V <sup>[3]</sup> | -                | ±5   | -                 | ±30  | µA   |
|                  |                           | suspend mode B port; V <sub>O</sub> = 0 V or V <sub>CCO</sub> ; V <sub>CC(A)</sub> = 0 V; V <sub>CC(B)</sub> = 3.6 V <sup>[3]</sup> | -                | ±5   | -                 | ±30  | µA   |
| I <sub>OFF</sub> | power-off leakage current | A port; V <sub>I</sub> or V <sub>O</sub> = 0 V to 3.6 V;<br>V <sub>CC(A)</sub> = 0 V; V <sub>CC(B)</sub> = 0.8 V to 3.6 V           | -                | ±5   | -                 | ±30  | µA   |
|                  |                           | B port; V <sub>I</sub> or V <sub>O</sub> = 0 V to 3.6 V;<br>V <sub>CC(B)</sub> = 0 V; V <sub>CC(A)</sub> = 0.8 V to 3.6 V           | -                | ±5   | -                 | ±30  | µA   |



| Symbol          | Parameter                 | Conditions                                                                                                                                                                                  | -40 °C to +85 °C |     | -40 °C to +125 °C |     | Unit          |
|-----------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----|-------------------|-----|---------------|
|                 |                           |                                                                                                                                                                                             | Min              | Max | Min               | Max |               |
| $I_{CC}$        | supply current            | A port; $V_I = 0\text{ V}$ or $V_{CCI}$ ; $I_O = 0\text{ A}$                                                                                                                                |                  |     |                   |     |               |
|                 |                           | $V_{CC(A)} = 0.8\text{ V to }3.6\text{ V};$<br>$V_{CC(B)} = 0.8\text{ V to }3.6\text{ V}$                                                                                                   | -                | 10  | -                 | 55  | $\mu\text{A}$ |
|                 |                           | $V_{CC(A)} = 1.1\text{ V to }3.6\text{ V};$<br>$V_{CC(B)} = 1.1\text{ V to }3.6\text{ V}$                                                                                                   | -                | 8   | -                 | 50  | $\mu\text{A}$ |
|                 |                           | $V_{CC(A)} = 3.6\text{ V}; V_{CC(B)} = 0\text{ V}$                                                                                                                                          | -                | 8   | -                 | 50  | $\mu\text{A}$ |
|                 |                           | $V_{CC(A)} = 0\text{ V}; V_{CC(B)} = 3.6\text{ V}$                                                                                                                                          | -2               | -   | -12               | -   | $\mu\text{A}$ |
|                 |                           | B port; $V_I = 0\text{ V}$ or $V_{CCI}$ ; $I_O = 0\text{ A}$                                                                                                                                |                  |     |                   |     |               |
|                 |                           | $V_{CC(A)} = 0.8\text{ V to }3.6\text{ V};$<br>$V_{CC(B)} = 0.8\text{ V to }3.6\text{ V}$                                                                                                   | -                | 10  | -                 | 55  | $\mu\text{A}$ |
|                 |                           | $V_{CC(A)} = 1.1\text{ V to }3.6\text{ V};$<br>$V_{CC(B)} = 1.1\text{ V to }3.6\text{ V}$                                                                                                   | -                | 8   | -                 | 50  | $\mu\text{A}$ |
|                 |                           | $V_{CC(A)} = 3.6\text{ V}; V_{CC(B)} = 0\text{ V}$                                                                                                                                          | -2               | -   | -12               | -   | $\mu\text{A}$ |
|                 |                           | $V_{CC(A)} = 0\text{ V}; V_{CC(B)} = 3.6\text{ V}$                                                                                                                                          | -                | 8   | -                 | 50  | $\mu\text{A}$ |
|                 |                           | A plus B port ( $I_{CC(A)} + I_{CC(B)}$ );<br>$I_O = 0\text{ A}; V_I = 0\text{ V or }V_{CCI};$<br>$V_{CC(A)} = 0.8\text{ V to }3.6\text{ V};$<br>$V_{CC(B)} = 0.8\text{ V to }3.6\text{ V}$ | -                | 20  | -                 | 70  | $\mu\text{A}$ |
|                 |                           | A plus B port ( $I_{CC(A)} + I_{CC(B)}$ );<br>$I_O = 0\text{ A}; V_I = 0\text{ V or }V_{CCI};$<br>$V_{CC(A)} = 1.1\text{ V to }3.6\text{ V};$<br>$V_{CC(B)} = 1.1\text{ V to }3.6\text{ V}$ | -                | 16  | -                 | 65  | $\mu\text{A}$ |
| $\Delta I_{CC}$ | additional supply current | $V_I = 3.0\text{ V}; V_{CC(A)} = V_{CC(B)} = 3.6\text{ V}$                                                                                                                                  | -                | 500 | -                 | 650 | $\mu\text{A}$ |

[1]  $V_{CCO}$  is the supply voltage associated with the output port.

[2]  $V_{CCI}$  is the supply voltage associated with the data input port.

[3] For I/O ports, the parameter  $I_{OZ}$  includes the input leakage current.

**Table 9. Typical total supply current ( $I_{CC(A)} + I_{CC(B)}$ )**

| $V_{CC(A)}$ | $V_{CC(B)}$ |       |       |       |       |       |       | Unit          |
|-------------|-------------|-------|-------|-------|-------|-------|-------|---------------|
|             | 0 V         | 0.8 V | 1.2 V | 1.5 V | 1.8 V | 2.5 V | 3.3 V |               |
| 0 V         | 0           | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | $\mu\text{A}$ |
| 0.8 V       | 0.1         | 0.1   | 0.1   | 0.1   | 0.1   | 0.3   | 1.6   | $\mu\text{A}$ |
| 1.2 V       | 0.1         | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.8   | $\mu\text{A}$ |
| 1.5 V       | 0.1         | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.4   | $\mu\text{A}$ |
| 1.8 V       | 0.1         | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.2   | $\mu\text{A}$ |
| 2.5 V       | 0.1         | 0.3   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | $\mu\text{A}$ |
| 3.3 V       | 0.1         | 1.6   | 0.8   | 0.4   | 0.2   | 0.1   | 0.1   | $\mu\text{A}$ |

## 11 Dynamic characteristics

**Table 10. Typical power dissipation capacitance at  $V_{CC(A)} = V_{CC(B)}$  and  $T_{amb} = 25\text{ °C}$  [1] [2]**

Voltages are referenced to GND (ground = 0 V).

| Symbol   | Parameter                     | Conditions                                    | $V_{CC(A)} = V_{CC(B)}$ |       |       |       |       |       | Unit |
|----------|-------------------------------|-----------------------------------------------|-------------------------|-------|-------|-------|-------|-------|------|
|          |                               |                                               | 0.8 V                   | 1.2 V | 1.5 V | 1.8 V | 2.5 V | 3.3 V |      |
| $C_{PD}$ | power dissipation capacitance | A port: (direction An to Bn); output enabled  | 0.2                     | 0.2   | 0.2   | 0.2   | 0.3   | 0.4   | pF   |
|          |                               | A port: (direction An to Bn); output disabled | 0.2                     | 0.2   | 0.2   | 0.2   | 0.3   | 0.4   | pF   |
|          |                               | A port: (direction Bn to An); output enabled  | 9.5                     | 9.7   | 9.8   | 9.9   | 10.7  | 11.9  | pF   |
|          |                               | A port: (direction Bn to An); output disabled | 0.6                     | 0.6   | 0.6   | 0.6   | 0.7   | 0.7   | pF   |
|          |                               | B port: (direction An to Bn); output enabled  | 9.5                     | 9.7   | 9.8   | 9.9   | 10.7  | 11.9  | pF   |
|          |                               | B port: (direction An to Bn); output disabled | 0.6                     | 0.6   | 0.6   | 0.6   | 0.7   | 0.7   | pF   |
|          |                               | B port: (direction Bn to An); output enabled  | 0.2                     | 0.2   | 0.2   | 0.2   | 0.3   | 0.4   | pF   |
|          |                               | B port: (direction Bn to An); output disabled | 0.2                     | 0.2   | 0.2   | 0.2   | 0.3   | 0.4   | pF   |

[1]  $C_{PD}$  is used to determine the dynamic power dissipation ( $P_D$  in  $\mu W$ ).

$$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o) \text{ where:}$$

$f_i$  = input frequency in MHz;

$f_o$  = output frequency in MHz;

$C_L$  = load capacitance in pF;

$V_{CC}$  = supply voltage in V;

$N$  = number of inputs switching;

$\Sigma(C_L \times V_{CC}^2 \times f_o)$  = sum of the outputs.

[2]  $f_i = 10\text{ MHz}$ ;  $V_I = \text{GND to } V_{CC}$ ;  $t_r = t_f = 1\text{ ns}$ ;  $C_L = 0\text{ pF}$ ;  $R_L = \infty\ \Omega$ .

**Table 11. Typical dynamic characteristics at  $V_{CC(A)} = 0.8\text{ V}$  and  $T_{amb} = 25\text{ °C}$  [1]**

Voltages are referenced to GND (ground = 0 V); for test circuit see [Figure 7](#); for waveforms see [Figure 5](#) and [Figure 6](#)

| Symbol    | Parameter         | Conditions            | $V_{CC(B)}$ |       |       |       |       |       | Unit |
|-----------|-------------------|-----------------------|-------------|-------|-------|-------|-------|-------|------|
|           |                   |                       | 0.8 V       | 1.2 V | 1.5 V | 1.8 V | 2.5 V | 3.3 V |      |
| $t_{pd}$  | propagation delay | An to Bn              | 14.5        | 7.3   | 6.5   | 6.2   | 5.9   | 6.0   | ns   |
|           |                   | Bn to An              | 14.5        | 12.7  | 12.4  | 12.3  | 12.1  | 12.0  | ns   |
| $t_{dis}$ | disable time      | $\overline{OE}$ to An | 14.3        | 14.3  | 14.3  | 14.3  | 14.3  | 14.3  | ns   |
|           |                   | $\overline{OE}$ to Bn | 17.0        | 9.9   | 9.0   | 9.4   | 9.0   | 9.7   | ns   |
| $t_{en}$  | enable time       | $\overline{OE}$ to An | 18.2        | 18.2  | 18.2  | 18.2  | 18.2  | 18.2  | ns   |
|           |                   | $\overline{OE}$ to Bn | 19.2        | 10.7  | 9.8   | 9.6   | 9.7   | 10.2  | ns   |

[1]  $t_{pd}$  is the same as  $t_{PLH}$  and  $t_{PHL}$ ;  $t_{dis}$  is the same as  $t_{PLZ}$  and  $t_{PHZ}$ ;  $t_{en}$  is the same as  $t_{PZL}$  and  $t_{PZH}$ .

**Table 12. Typical dynamic characteristics at  $V_{CC(B)} = 0.8\text{ V}$  and  $T_{amb} = 25\text{ °C}$  [1]**

Voltages are referenced to GND (ground = 0 V); for test circuit see [Figure 7](#); for waveforms see [Figure 5](#) and [Figure 6](#)

| Symbol    | Parameter         | Conditions            | $V_{CC(A)}$ |       |       |       |       |       | Unit |
|-----------|-------------------|-----------------------|-------------|-------|-------|-------|-------|-------|------|
|           |                   |                       | 0.8 V       | 1.2 V | 1.5 V | 1.8 V | 2.5 V | 3.3 V |      |
| $t_{pd}$  | propagation delay | An to Bn              | 14.5        | 12.7  | 12.4  | 12.3  | 12.1  | 12.0  | ns   |
|           |                   | Bn to An              | 14.5        | 7.3   | 6.5   | 6.2   | 5.9   | 6.0   | ns   |
| $t_{dis}$ | disable time      | $\overline{OE}$ to An | 14.3        | 5.5   | 4.1   | 4.0   | 3.0   | 3.5   | ns   |
|           |                   | $\overline{OE}$ to Bn | 17.0        | 13.8  | 13.4  | 13.1  | 12.9  | 12.7  | ns   |
| $t_{en}$  | enable time       | $\overline{OE}$ to An | 18.2        | 5.6   | 4.0   | 3.2   | 2.4   | 2.2   | ns   |
|           |                   | $\overline{OE}$ to Bn | 19.2        | 14.6  | 14.1  | 13.9  | 13.7  | 13.6  | ns   |

[1]  $t_{pd}$  is the same as  $t_{PLH}$  and  $t_{PHL}$ ;  $t_{dis}$  is the same as  $t_{PLZ}$  and  $t_{PHZ}$ ;  $t_{en}$  is the same as  $t_{PZL}$  and  $t_{PZH}$ .

**Table 13. Dynamic characteristics for temperature range -40 °C to +85 °C <sup>[1]</sup>**

Voltages are referenced to GND (ground = 0 V); for test circuit see [Figure 7](#); for waveforms see [Figure 5](#) and [Figure 6](#)

| Symbol                                | Parameter         | Conditions            | V <sub>CC(B)</sub> |      |              |      |               |      |              |      |              |      | Unit |
|---------------------------------------|-------------------|-----------------------|--------------------|------|--------------|------|---------------|------|--------------|------|--------------|------|------|
|                                       |                   |                       | 1.2 V ±0.1 V       |      | 1.5 V ±0.1 V |      | 1.8 V ±0.15 V |      | 2.5 V ±0.2 V |      | 3.3 V ±0.3 V |      |      |
|                                       |                   |                       | Min                | Max  | Min          | Max  | Min           | Max  | Min          | Max  | Min          | Max  |      |
| V <sub>CC(A)</sub> = 1.1 V to 1.3 V   |                   |                       |                    |      |              |      |               |      |              |      |              |      |      |
| t <sub>pd</sub>                       | propagation delay | An to Bn              | 2.0                | 10.5 | 1.3          | 7.8  | 1.2           | 6.9  | 1.0          | 5.9  | 0.8          | 5.7  | ns   |
|                                       |                   | Bn to An              | 2.0                | 10.5 | 1.5          | 9.9  | 1.5           | 9.7  | 1.4          | 9.4  | 1.4          | 9.3  | ns   |
| t <sub>dis</sub>                      | disable time      | $\overline{OE}$ to An | 2.0                | 10.0 | 2.0          | 10.0 | 2.0           | 10.0 | 2.0          | 10.0 | 2.0          | 10.0 | ns   |
|                                       |                   | $\overline{OE}$ to Bn | 2.0                | 11.1 | 2.0          | 8.6  | 1.0           | 8.0  | 0.7          | 7.0  | 1.0          | 8.0  | ns   |
| t <sub>en</sub>                       | enable time       | $\overline{OE}$ to An | 2.0                | 13.5 | 2.0          | 13.5 | 2.0           | 13.5 | 2.0          | 13.5 | 2.0          | 13.5 | ns   |
|                                       |                   | $\overline{OE}$ to Bn | 2.0                | 15.0 | 2.0          | 11.0 | 2.0           | 9.4  | 1.0          | 7.8  | 1.0          | 7.4  | ns   |
| V <sub>CC(A)</sub> = 1.4 V to 1.6 V   |                   |                       |                    |      |              |      |               |      |              |      |              |      |      |
| t <sub>pd</sub>                       | propagation delay | An to Bn              | 1.5                | 9.9  | 1.0          | 7.1  | 1.0           | 6.0  | 0.5          | 4.8  | 0.5          | 4.3  | ns   |
|                                       |                   | Bn to An              | 1.3                | 7.8  | 1.0          | 7.1  | 0.9           | 6.9  | 0.8          | 6.6  | 0.6          | 6.5  | ns   |
| t <sub>dis</sub>                      | disable time      | $\overline{OE}$ to An | 1.0                | 6.0  | 1.0          | 6.0  | 1.0           | 6.0  | 1.0          | 6.0  | 1.0          | 6.0  | ns   |
|                                       |                   | $\overline{OE}$ to Bn | 2.0                | 10.2 | 1.5          | 7.5  | 0.9           | 7.2  | 0.4          | 6.2  | 0.4          | 6.1  | ns   |
| t <sub>en</sub>                       | enable time       | $\overline{OE}$ to An | 1.0                | 7.5  | 1.0          | 7.5  | 1.0           | 7.5  | 1.0          | 7.5  | 1.0          | 7.5  | ns   |
|                                       |                   | $\overline{OE}$ to Bn | 2.0                | 14.4 | 1.4          | 7.9  | 1.3           | 7.7  | 1.1          | 6.4  | 1.1          | 5.6  | ns   |
| V <sub>CC(A)</sub> = 1.65 V to 1.95 V |                   |                       |                    |      |              |      |               |      |              |      |              |      |      |
| t <sub>pd</sub>                       | propagation delay | An to Bn              | 1.5                | 9.7  | 0.9          | 6.9  | 0.8           | 5.7  | 0.5          | 4.5  | 0.3          | 4.0  | ns   |
|                                       |                   | Bn to An              | 1.2                | 6.9  | 1.0          | 6.0  | 0.8           | 5.7  | 0.5          | 5.5  | 0.5          | 5.3  | ns   |
| t <sub>dis</sub>                      | disable time      | $\overline{OE}$ to An | 0.5                | 5.7  | 0.5          | 5.7  | 0.5           | 5.7  | 0.5          | 5.7  | 0.5          | 5.7  | ns   |
|                                       |                   | $\overline{OE}$ to Bn | 2.0                | 9.9  | 1.5          | 7.0  | 0.8           | 6.9  | 0.2          | 5.8  | 0.2          | 5.9  | ns   |
| t <sub>en</sub>                       | enable time       | $\overline{OE}$ to An | 1.0                | 6.7  | 1.0          | 6.7  | 1.0           | 6.7  | 1.0          | 6.7  | 1.0          | 6.7  | ns   |
|                                       |                   | $\overline{OE}$ to Bn | 1.5                | 13.9 | 1.2          | 7.2  | 1.2           | 6.9  | 0.8          | 5.4  | 0.6          | 5.0  | ns   |
| V <sub>CC(A)</sub> = 2.3 V to 2.7 V   |                   |                       |                    |      |              |      |               |      |              |      |              |      |      |
| t <sub>pd</sub>                       | propagation delay | An to Bn              | 1.4                | 9.4  | 0.8          | 6.6  | 0.5           | 5.5  | 0.4          | 4.2  | 0.2          | 3.7  | ns   |
|                                       |                   | Bn to An              | 1.0                | 5.9  | 0.5          | 4.8  | 0.5           | 4.5  | 0.4          | 4.2  | 0.3          | 3.9  | ns   |
| t <sub>dis</sub>                      | disable time      | $\overline{OE}$ to An | 0.2                | 4.0  | 0.2          | 4.0  | 0.2           | 4.0  | 0.2          | 4.0  | 0.2          | 4.0  | ns   |
|                                       |                   | $\overline{OE}$ to Bn | 2.0                | 9.3  | 1.5          | 6.7  | 0.7           | 6.3  | 0.2          | 5.0  | 0.2          | 5.7  | ns   |
| t <sub>en</sub>                       | enable time       | $\overline{OE}$ to An | 0.6                | 4.5  | 0.6          | 4.5  | 0.6           | 4.5  | 0.6          | 4.5  | 0.6          | 4.5  | ns   |
|                                       |                   | $\overline{OE}$ to Bn | 1.5                | 13.6 | 1.0          | 6.8  | 1.0           | 6.0  | 0.8          | 4.6  | 0.6          | 4.2  | ns   |

| Symbol                              | Parameter         | Conditions                   | V <sub>CC(B)</sub> |      |              |     |               |     |              |     |              |     | Unit |
|-------------------------------------|-------------------|------------------------------|--------------------|------|--------------|-----|---------------|-----|--------------|-----|--------------|-----|------|
|                                     |                   |                              | 1.2 V ±0.1 V       |      | 1.5 V ±0.1 V |     | 1.8 V ±0.15 V |     | 2.5 V ±0.2 V |     | 3.3 V ±0.3 V |     |      |
|                                     |                   |                              | Min                | Max  | Min          | Max | Min           | Max | Min          | Max | Min          | Max |      |
| V <sub>CC(A)</sub> = 3.0 V to 3.6 V |                   |                              |                    |      |              |     |               |     |              |     |              |     |      |
| t <sub>pd</sub>                     | propagation delay | An to Bn                     | 1.4                | 9.3  | 0.6          | 6.5 | 0.5           | 5.3 | 0.3          | 3.9 | 0.2          | 3.5 | ns   |
|                                     |                   | Bn to An                     | 0.8                | 5.7  | 0.5          | 4.3 | 0.3           | 4.0 | 0.2          | 3.7 | 0.2          | 3.5 | ns   |
| t <sub>dis</sub>                    | disable time      | $\overline{\text{OE}}$ to An | 0.2                | 4.5  | 0.2          | 4.5 | 0.2           | 4.5 | 0.2          | 4.5 | 0.2          | 4.5 | ns   |
|                                     |                   | $\overline{\text{OE}}$ to Bn | 2.0                | 9.0  | 1.5          | 6.4 | 0.7           | 6.1 | 0.2          | 4.8 | 0.2          | 5.6 | ns   |
| t <sub>en</sub>                     | enable time       | $\overline{\text{OE}}$ to An | 0.5                | 4.0  | 0.5          | 4.0 | 0.5           | 4.0 | 0.5          | 4.0 | 0.5          | 4.0 | ns   |
|                                     |                   | $\overline{\text{OE}}$ to Bn | 1.5                | 13.4 | 1.0          | 6.7 | 1.0           | 5.9 | 0.7          | 4.4 | 0.5          | 4.0 | ns   |

[1] t<sub>pd</sub> is the same as t<sub>PLH</sub> and t<sub>PHL</sub>; t<sub>dis</sub> is the same as t<sub>PLZ</sub> and t<sub>PHZ</sub>; t<sub>en</sub> is the same as t<sub>PZL</sub> and t<sub>PZH</sub>.

**Table 14. Dynamic characteristics for temperature range -40 °C to +125 °C [1]**

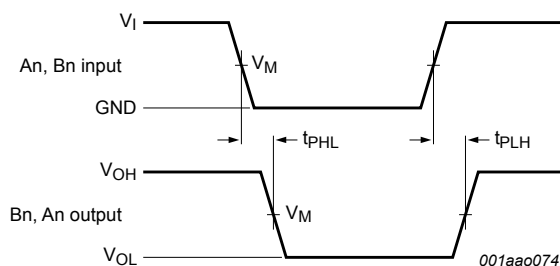
Voltages are referenced to GND (ground = 0 V); for test circuit see [Figure 7](#); for waveforms see [Figure 5](#) and [Figure 6](#)

| Symbol                              | Parameter         | Conditions                   | V <sub>CC(B)</sub> |      |              |      |               |      |              |      |              |      | Unit |
|-------------------------------------|-------------------|------------------------------|--------------------|------|--------------|------|---------------|------|--------------|------|--------------|------|------|
|                                     |                   |                              | 1.2 V ±0.1 V       |      | 1.5 V ±0.1 V |      | 1.8 V ±0.15 V |      | 2.5 V ±0.2 V |      | 3.3 V ±0.3 V |      |      |
|                                     |                   |                              | Min                | Max  | Min          | Max  | Min           | Max  | Min          | Max  | Min          | Max  |      |
| V <sub>CC(A)</sub> = 1.1 V to 1.3 V |                   |                              |                    |      |              |      |               |      |              |      |              |      |      |
| t <sub>pd</sub>                     | propagation delay | An to Bn                     | 2.0                | 12.1 | 1.3          | 9.0  | 1.2           | 8.0  | 1.0          | 6.8  | 0.8          | 6.6  | ns   |
|                                     |                   | Bn to An                     | 2.0                | 12.1 | 1.5          | 11.4 | 1.5           | 11.2 | 1.4          | 10.9 | 1.4          | 10.7 | ns   |
| t <sub>dis</sub>                    | disable time      | $\overline{\text{OE}}$ to An | 2.0                | 11.5 | 2.0          | 11.5 | 2.0           | 11.5 | 2.0          | 11.5 | 2.0          | 11.5 | ns   |
|                                     |                   | $\overline{\text{OE}}$ to Bn | 2.0                | 12.8 | 2.0          | 9.9  | 1.0           | 9.2  | 0.7          | 8.1  | 1.0          | 9.2  | ns   |
| t <sub>en</sub>                     | enable time       | $\overline{\text{OE}}$ to An | 2.0                | 15.6 | 2.0          | 15.6 | 2.0           | 15.6 | 2.0          | 15.6 | 2.0          | 15.6 | ns   |
|                                     |                   | $\overline{\text{OE}}$ to Bn | 2.0                | 17.3 | 2.0          | 12.7 | 2.0           | 10.9 | 1.0          | 9.0  | 1.0          | 8.6  | ns   |
| V <sub>CC(A)</sub> = 1.4 V to 1.6 V |                   |                              |                    |      |              |      |               |      |              |      |              |      |      |
| t <sub>pd</sub>                     | propagation delay | An to Bn                     | 1.5                | 11.4 | 1.0          | 8.2  | 1.0           | 6.9  | 0.5          | 5.6  | 0.5          | 5.0  | ns   |
|                                     |                   | Bn to An                     | 1.3                | 9.0  | 1.0          | 8.2  | 0.9           | 8.0  | 0.8          | 7.6  | 0.6          | 7.5  | ns   |
| t <sub>dis</sub>                    | disable time      | $\overline{\text{OE}}$ to An | 1.0                | 6.9  | 1.0          | 6.9  | 1.0           | 6.9  | 1.0          | 6.9  | 1.0          | 6.9  | ns   |
|                                     |                   | $\overline{\text{OE}}$ to Bn | 2.0                | 11.8 | 1.5          | 8.7  | 0.9           | 8.3  | 0.4          | 7.2  | 0.4          | 7.1  | ns   |
| t <sub>en</sub>                     | enable time       | $\overline{\text{OE}}$ to An | 1.0                | 8.7  | 1.0          | 8.7  | 1.0           | 8.7  | 1.0          | 8.7  | 1.0          | 8.7  | ns   |
|                                     |                   | $\overline{\text{OE}}$ to Bn | 2.0                | 16.6 | 1.4          | 9.1  | 1.3           | 8.9  | 1.1          | 7.4  | 1.1          | 6.5  | ns   |

| Symbol                                | Parameter         | Conditions | V <sub>CC(B)</sub> |      |              |     |               |     |              |     |              |     | Unit |
|---------------------------------------|-------------------|------------|--------------------|------|--------------|-----|---------------|-----|--------------|-----|--------------|-----|------|
|                                       |                   |            | 1.2 V ±0.1 V       |      | 1.5 V ±0.1 V |     | 1.8 V ±0.15 V |     | 2.5 V ±0.2 V |     | 3.3 V ±0.3 V |     |      |
|                                       |                   |            | Min                | Max  | Min          | Max | Min           | Max | Min          | Max | Min          | Max |      |
| V <sub>CC(A)</sub> = 1.65 V to 1.95 V |                   |            |                    |      |              |     |               |     |              |     |              |     |      |
| t <sub>pd</sub>                       | propagation delay | An to Bn   | 1.5                | 11.2 | 0.9          | 8.0 | 0.8           | 6.6 | 0.5          | 5.2 | 0.3          | 4.6 | ns   |
|                                       |                   | Bn to An   | 1.2                | 8.0  | 1.0          | 6.9 | 0.8           | 6.6 | 0.5          | 6.4 | 0.5          | 6.1 | ns   |
| t <sub>dis</sub>                      | disable time      | OE to An   | 0.5                | 6.6  | 0.5          | 6.6 | 0.5           | 6.6 | 0.5          | 6.6 | 0.5          | 6.6 | ns   |
|                                       |                   | OE to Bn   | 2.0                | 11.4 | 1.5          | 8.1 | 0.8           | 8.0 | 0.2          | 6.7 | 0.2          | 6.8 | ns   |
| t <sub>en</sub>                       | enable time       | OE to An   | 1.0                | 7.8  | 1.0          | 7.8 | 1.0           | 7.8 | 1.0          | 7.8 | 1.0          | 7.8 | ns   |
|                                       |                   | OE to Bn   | 1.5                | 16.0 | 1.2          | 8.3 | 1.2           | 8.0 | 0.8          | 6.3 | 0.6          | 5.8 | ns   |
| V <sub>CC(A)</sub> = 2.3 V to 2.7 V   |                   |            |                    |      |              |     |               |     |              |     |              |     |      |
| t <sub>pd</sub>                       | propagation delay | An to Bn   | 1.4                | 10.9 | 0.8          | 7.6 | 0.5           | 6.4 | 0.4          | 4.9 | 0.2          | 4.3 | ns   |
|                                       |                   | Bn to An   | 1.0                | 6.8  | 0.5          | 5.6 | 0.5           | 5.2 | 0.4          | 4.9 | 0.3          | 4.5 | ns   |
| t <sub>dis</sub>                      | disable time      | OE to An   | 0.2                | 4.6  | 0.2          | 4.6 | 0.2           | 4.6 | 0.2          | 4.6 | 0.2          | 4.6 | ns   |
|                                       |                   | OE to Bn   | 2.0                | 10.7 | 1.5          | 7.8 | 0.7           | 7.3 | 0.2          | 5.8 | 0.2          | 6.6 | ns   |
| t <sub>en</sub>                       | enable time       | OE to An   | 0.6                | 5.2  | 0.6          | 5.2 | 0.6           | 5.2 | 0.6          | 5.2 | 0.6          | 5.2 | ns   |
|                                       |                   | OE to Bn   | 1.5                | 15.7 | 1.0          | 7.9 | 1.0           | 6.9 | 0.8          | 5.3 | 0.6          | 4.9 | ns   |
| V <sub>CC(A)</sub> = 3.0 V to 3.6 V   |                   |            |                    |      |              |     |               |     |              |     |              |     |      |
| t <sub>pd</sub>                       | propagation delay | An to Bn   | 1.4                | 10.7 | 0.6          | 7.5 | 0.5           | 6.1 | 0.3          | 4.5 | 0.2          | 4.1 | ns   |
|                                       |                   | Bn to An   | 0.8                | 6.6  | 0.5          | 5.0 | 0.3           | 4.6 | 0.2          | 4.3 | 0.2          | 4.1 | ns   |
| t <sub>dis</sub>                      | disable time      | OE to An   | 0.2                | 5.2  | 0.2          | 5.2 | 0.2           | 5.2 | 0.2          | 5.2 | 0.2          | 5.2 | ns   |
|                                       |                   | OE to Bn   | 2.0                | 10.4 | 1.5          | 7.4 | 0.7           | 7.1 | 0.2          | 5.6 | 0.2          | 6.5 | ns   |
| t <sub>en</sub>                       | enable time       | OE to An   | 0.5                | 4.6  | 0.5          | 4.6 | 0.5           | 4.6 | 0.5          | 4.6 | 0.5          | 4.6 | ns   |
|                                       |                   | OE to Bn   | 1.5                | 15.5 | 1.0          | 7.8 | 1.0           | 6.8 | 0.7          | 5.1 | 0.5          | 4.6 | ns   |

[1]  $t_{pd}$  is the same as  $t_{PLH}$  and  $t_{PHL}$ ;  $t_{dis}$  is the same as  $t_{PLZ}$  and  $t_{PHZ}$ ;  $t_{en}$  is the same as  $t_{PZL}$  and  $t_{PZH}$ .

## 11.1 Waveforms and test circuit



Measurement points are given in [Table 15](#).

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

**Figure 5. The data input (An, Bn) to output (Bn, An) propagation delay times**

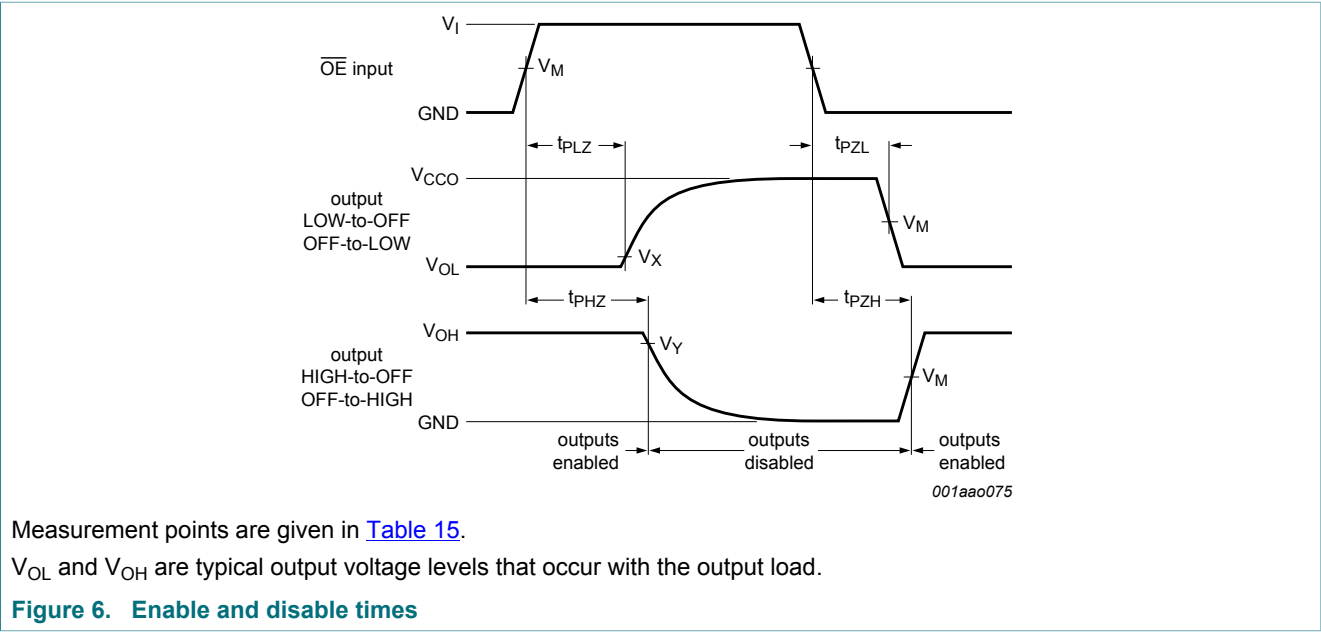
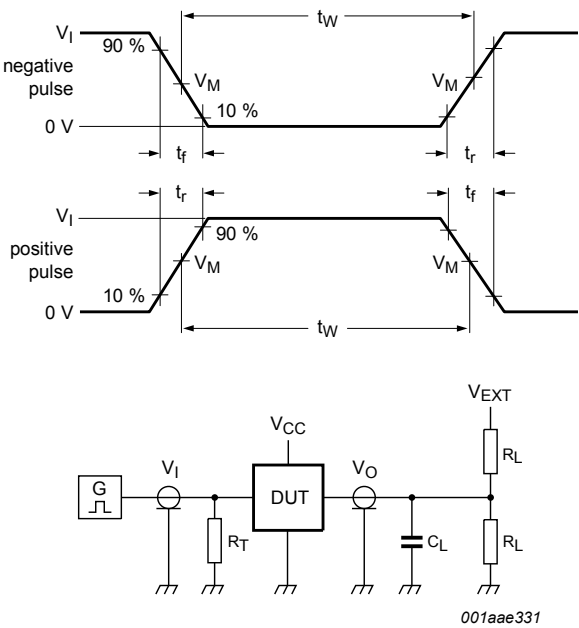


Table 15. Measurement points

| Supply voltage         | Input <sup>[1]</sup> | Output <sup>[2]</sup> |                          |                          |
|------------------------|----------------------|-----------------------|--------------------------|--------------------------|
| $V_{CC(A)}, V_{CC(B)}$ | $V_M$                | $V_M$                 | $V_X$                    | $V_Y$                    |
| 0.8 V to 1.6 V         | $0.5V_{CCI}$         | $0.5V_{CCO}$          | $V_{OL} + 0.1\text{ V}$  | $V_{OH} - 0.1\text{ V}$  |
| 1.65 V to 2.7 V        | $0.5V_{CCI}$         | $0.5V_{CCO}$          | $V_{OL} + 0.15\text{ V}$ | $V_{OH} - 0.15\text{ V}$ |
| 3.0 V to 3.6 V         | $0.5V_{CCI}$         | $0.5V_{CCO}$          | $V_{OL} + 0.3\text{ V}$  | $V_{OH} - 0.3\text{ V}$  |

[1]  $V_{CCI}$  is the supply voltage associated with the data input port.  
[2]  $V_{CCO}$  is the supply voltage associated with the output port.



Test data is given in [Table 16](#).  
 $R_L$  = Load resistance.  
 $C_L$  = Load capacitance including jig and probe capacitance.  
 $R_T$  = Termination resistance should be equal to the output impedance  $Z_o$  of the pulse generator.  
 $V_{EXT}$  = External voltage for measuring switching times.

Figure 7. Test circuit for measuring switching times

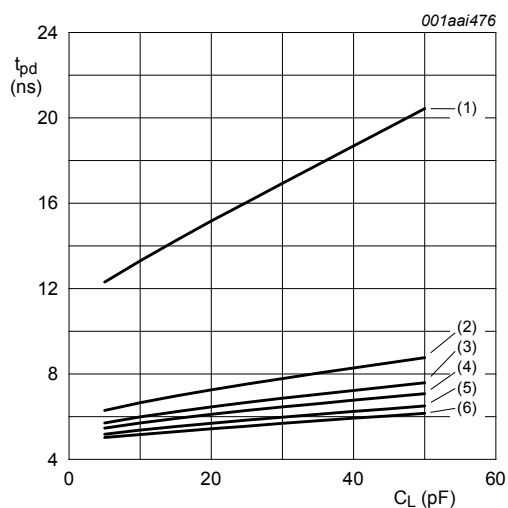
Table 16. Test data

| Supply voltage         | Input     |                         | Load  |              | $V_{EXT}$          |                    |                        |
|------------------------|-----------|-------------------------|-------|--------------|--------------------|--------------------|------------------------|
| $V_{CC(A)}, V_{CC(B)}$ | $V_I$ [1] | $\Delta t/\Delta V$ [2] | $C_L$ | $R_L$        | $t_{PLH}, t_{PHL}$ | $t_{PZH}, t_{PHZ}$ | $t_{PZL}, t_{PLZ}$ [3] |
| 0.8 V to 1.6 V         | $V_{CCI}$ | $\leq 1.0 \text{ ns/V}$ | 15 pF | 2 k $\Omega$ | open               | GND                | $2V_{CCO}$             |
| 1.65 V to 2.7 V        | $V_{CCI}$ | $\leq 1.0 \text{ ns/V}$ | 15 pF | 2 k $\Omega$ | open               | GND                | $2V_{CCO}$             |
| 3.0 V to 3.6 V         | $V_{CCI}$ | $\leq 1.0 \text{ ns/V}$ | 15 pF | 2 k $\Omega$ | open               | GND                | $2V_{CCO}$             |

[1]  $V_{CCI}$  is the supply voltage associated with the data input port.  
[2]  $dV/dt \geq 1.0 \text{ V/ns}$   
[3]  $V_{CCO}$  is the supply voltage associated with the output port.

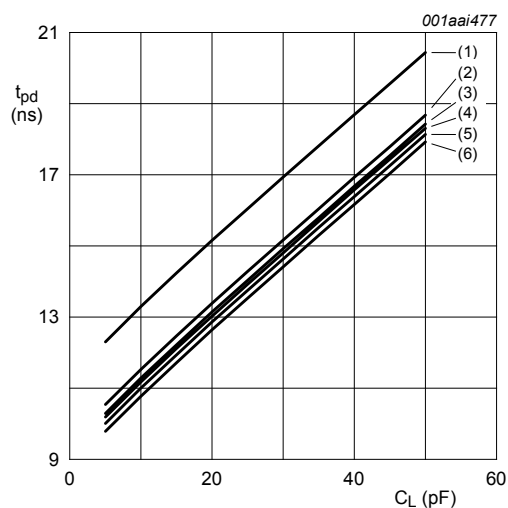


## 11.2 Typical propagation delay characteristics



a. Propagation delay (A to B);  $V_{CC(A)} = 0.8 \text{ V}$

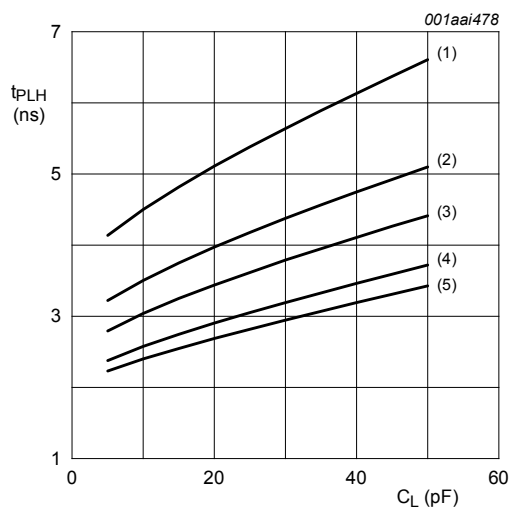
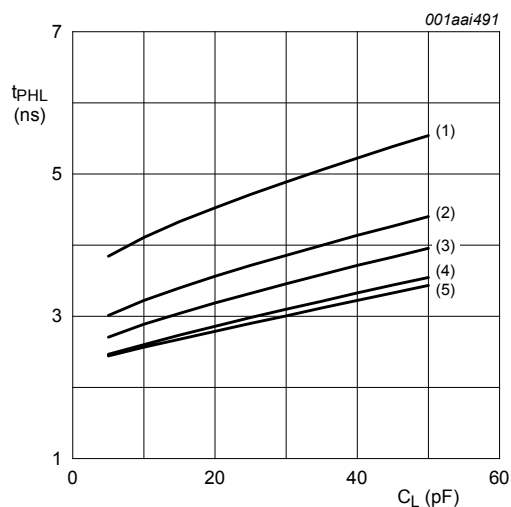
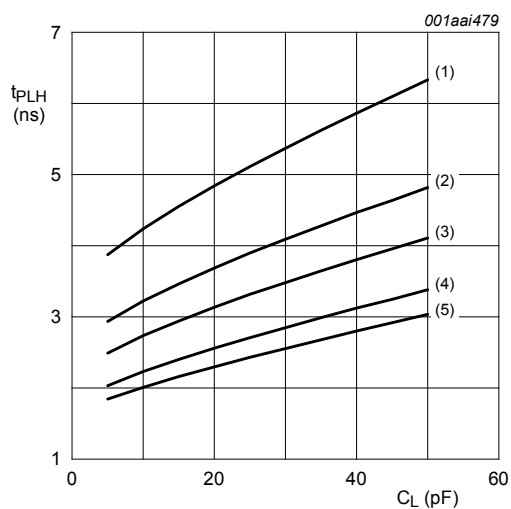
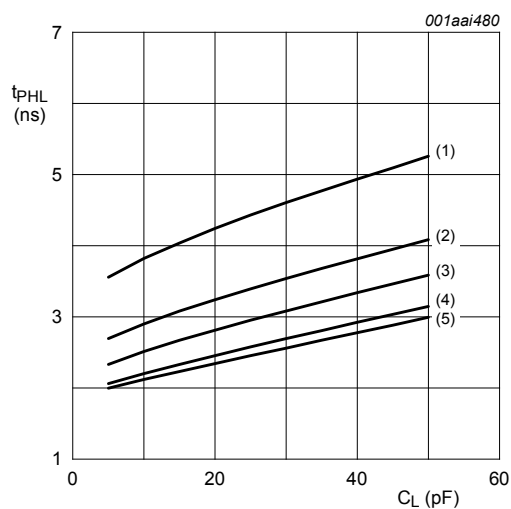
- (1)  $V_{CC(B)} = 0.8 \text{ V}$
- (2)  $V_{CC(B)} = 1.2 \text{ V}$
- (3)  $V_{CC(B)} = 1.5 \text{ V}$
- (4)  $V_{CC(B)} = 1.8 \text{ V}$
- (5)  $V_{CC(B)} = 2.5 \text{ V}$
- (6)  $V_{CC(B)} = 3.3 \text{ V}$



b. Propagation delay (A to B);  $V_{CC(B)} = 0.8 \text{ V}$

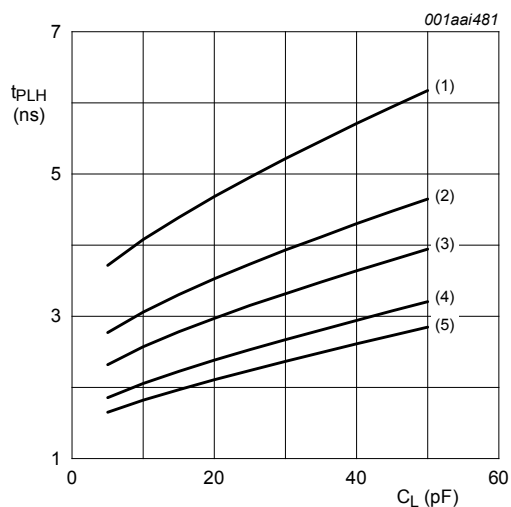
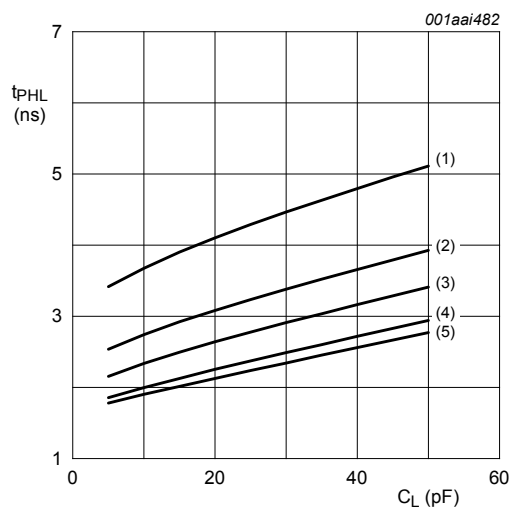
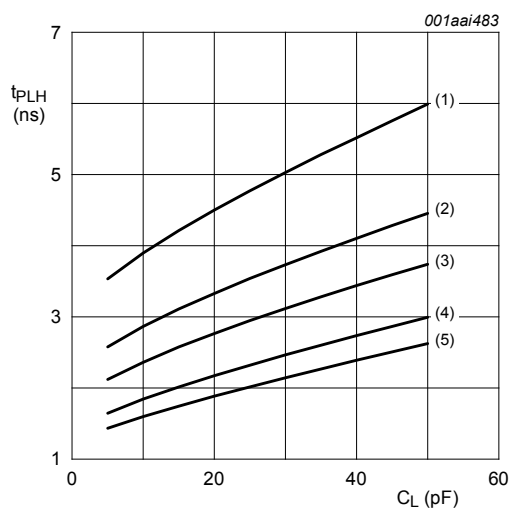
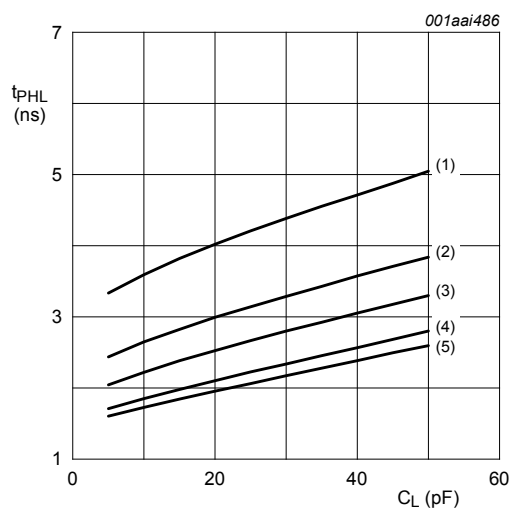
- (1)  $V_{CC(A)} = 0.8 \text{ V}$
- (2)  $V_{CC(A)} = 1.2 \text{ V}$
- (3)  $V_{CC(A)} = 1.5 \text{ V}$
- (4)  $V_{CC(A)} = 1.8 \text{ V}$
- (5)  $V_{CC(A)} = 2.5 \text{ V}$
- (6)  $V_{CC(A)} = 3.3 \text{ V}$

**Figure 8. Typical propagation delay versus load capacitance;  $T_{amb} = 25^\circ\text{C}$**

a. LOW to HIGH propagation delay (A to B);  $V_{CC(A)} = 1.2$  Vb. HIGH to LOW propagation delay (A to B);  $V_{CC(A)} = 1.2$  Vc. LOW to HIGH propagation delay (A to B);  $V_{CC(A)} = 1.5$  Vd. HIGH to LOW propagation delay (A to B);  $V_{CC(A)} = 1.5$  V

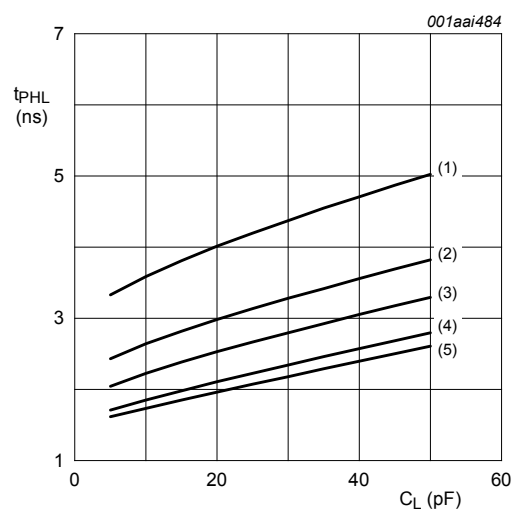
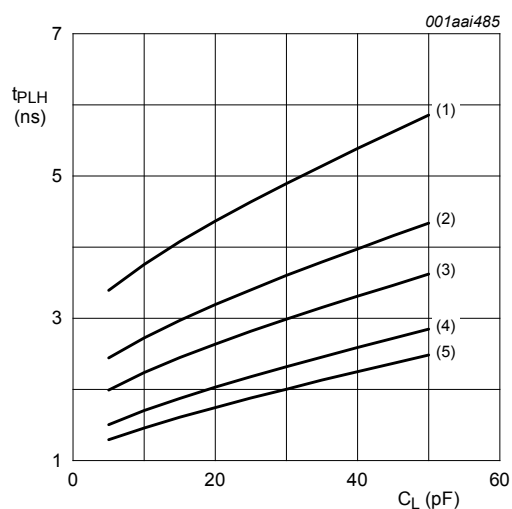
- (1)  $V_{CC(B)} = 1.2$  V  
 (2)  $V_{CC(B)} = 1.5$  V  
 (3)  $V_{CC(B)} = 1.8$  V  
 (4)  $V_{CC(B)} = 2.5$  V  
 (5)  $V_{CC(B)} = 3.3$  V

**Figure 9. Typical propagation delay versus load capacitance;  $T_{amb} = 25$  °C**

a. LOW to HIGH propagation delay (A to B);  $V_{CC(A)} = 1.8 \text{ V}$ b. HIGH to LOW propagation delay (A to B);  $V_{CC(A)} = 1.8 \text{ V}$ c. LOW to HIGH propagation delay (A to B);  $V_{CC(A)} = 2.5 \text{ V}$ d. HIGH to LOW propagation delay (A to B);  $V_{CC(A)} = 2.5 \text{ V}$ 

- (1)  $V_{CC(B)} = 1.2 \text{ V}$
- (2)  $V_{CC(B)} = 1.5 \text{ V}$
- (3)  $V_{CC(B)} = 1.8 \text{ V}$
- (4)  $V_{CC(B)} = 2.5 \text{ V}$
- (5)  $V_{CC(B)} = 3.3 \text{ V}$

**Figure 10. Typical propagation delay versus load capacitance;  $T_{amb} = 25^\circ\text{C}$**



a. LOW to HIGH propagation delay (A to B);  $V_{CC(A)} = 3.3 \text{ V}$

b. HIGH to LOW propagation delay (A to B);  $V_{CC(A)} = 3.3 \text{ V}$

(1)  $V_{CC(B)} = 1.2 \text{ V}$

(2)  $V_{CC(B)} = 1.5 \text{ V}$

(3)  $V_{CC(B)} = 1.8 \text{ V}$

(4)  $V_{CC(B)} = 2.5 \text{ V}$

(5)  $V_{CC(B)} = 3.3 \text{ V}$

**Figure 11. Typical propagation delay versus load capacitance;  $T_{amb} = 25 \text{ }^{\circ}\text{C}$**

12 Package outline

DHVQFN16: plastic dual in-line compatible thermal enhanced very thin quad flat package; no leads; 16 terminals; body 2.5 x 3.5 x 0.85 mm SOT763-1

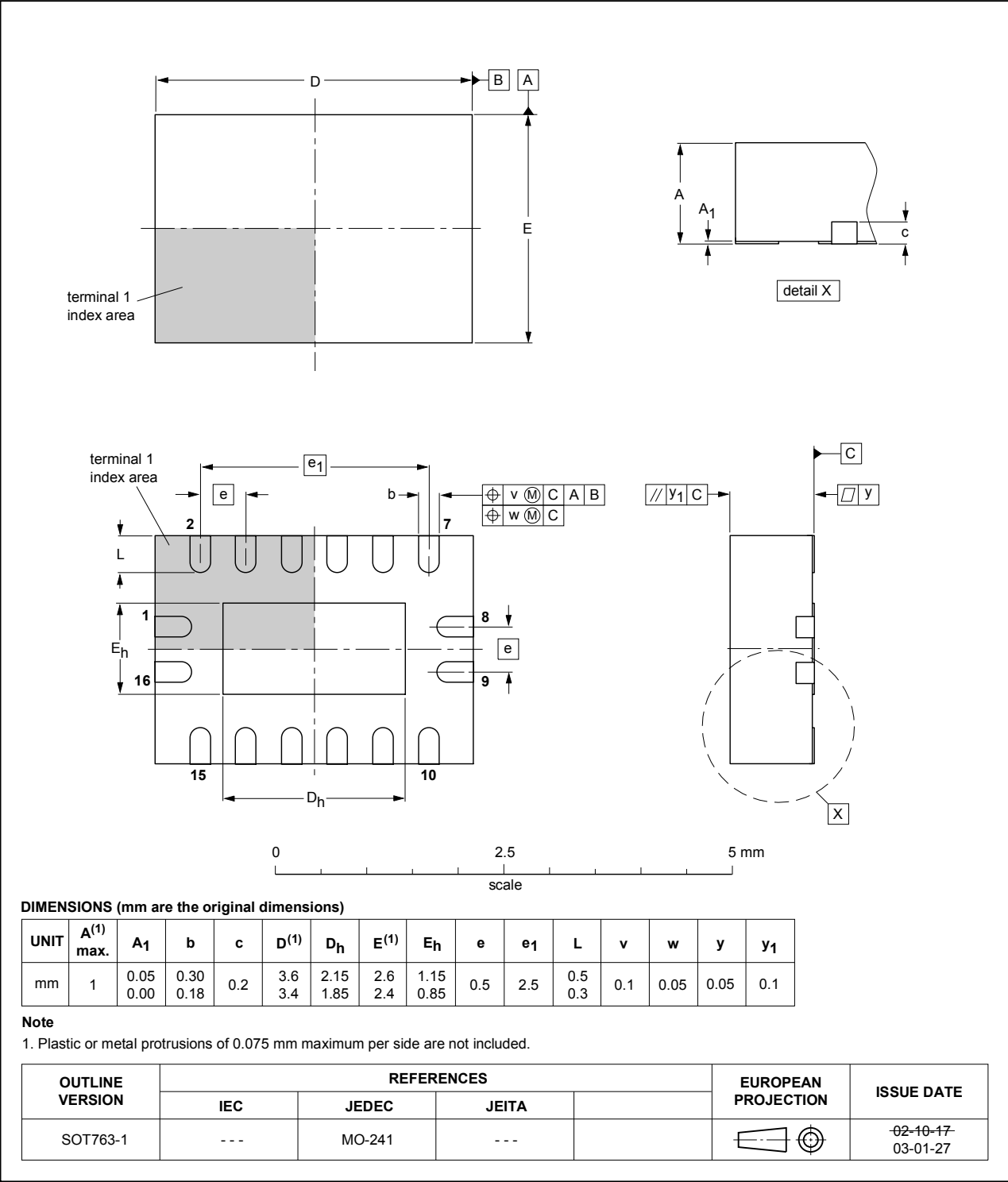


Figure 12. Package outline SOT763-1 (DHVQFN16)

XQFN16: plastic, extremely thin quad flat package; no leads;  
16 terminals; body 1.80 x 2.60 x 0.50 mm

SOT1161-1

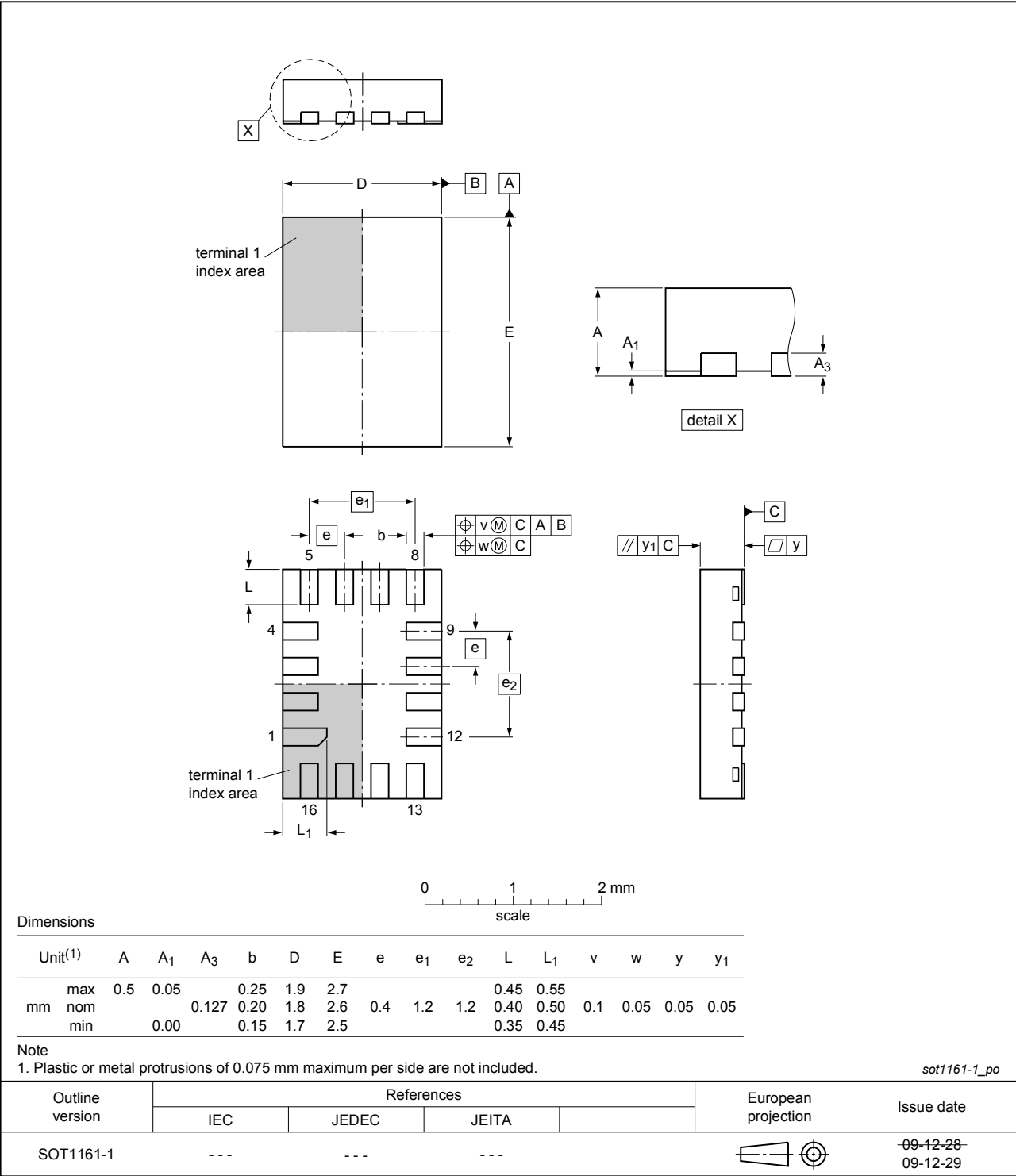


Figure 13. Package outline SOT1161-1 (XQFN16)

## 13 Abbreviations

Table 17. Abbreviations

| Acronym | Description             |
|---------|-------------------------|
| CDM     | Charged Device Model    |
| DUT     | Device Under Test       |
| ESD     | ElectroStatic Discharge |
| HBM     | Human Body Model        |

## 14 Revision history

Table 18. Revision history

| Document ID    | Release date | Data sheet status  | Change notice | Supersedes |
|----------------|--------------|--------------------|---------------|------------|
| 74AVC4T774 v.1 | 20170925     | Product data sheet | -             | -          |

## 15 Legal information

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| Document status <sup>[1][2]</sup> | Product status <sup>[3]</sup> | Definition                                                                            |
|-----------------------------------|-------------------------------|---------------------------------------------------------------------------------------|
| Objective [short] data sheet      | Development                   | This document contains data from the objective specification for product development. |
| Preliminary [short] data sheet    | Qualification                 | This document contains data from the preliminary specification.                       |
| Product [short] data sheet        | Production                    | This document contains the product specification.                                     |

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[2] The term 'short data sheet' is explained in section "Definitions".

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